Chapter 8

Criterion 8: Program Outcomes (POs) and Assessment

8.1 Program outcome (PO) statements and their mapping with the PEOs

Graduates of the B.Sc. in Computer Science and Engineering (CSE) program are expected to achieve the Program Outcomes (POs) listed in Table 8.1 by the time of graduation. The official approval document for these Program Outcomes is provided in section G.1.

Table 8.1: Program Outcomes

POs	Description
PO1	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to the solution of complex engineering problems.
PO2	Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (WK1 to WK4)
PO3	Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (WK5)
PO4	Conduct investigations of complex problems using research-based knowledge (WK8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (WK6)
PO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (WK7)
PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (WK7)
PO8	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (WK7)
PO9	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

POs	Description
PO11	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

The PO statements and their mapping with the PEOs is presented in Table 8.2. The cells marked with ✓ denote correlation between POs and PEOs.

Table 8.2: Mapping between PO statements and PEOs

Program Outcomes (POs)	**Program Educational Objectives (PEOs)**			
	PEO1	**PEO2**	**PEO3**	**PEO4**
PO1: Engineering Knowledge	✓			
PO2: Problem Analysis	✓			
PO3: Design/Development of Solutions	✓			
PO4: Investigation	✓			
PO5: Modern Tool Usage	✓			
PO6: The Engineer and Society		✓		
PO7: Environment and Sustainability		✓		
PO8: Ethics		✓		
PO9: Individual Work and Teamwork				✓
PO10: Communication				✓
PO11: Project Management and Finance				✓
PO12: Life-Long Learning			✓	

Justification for Mapping

This subsection explains the rationale behind the relationship between each Program Outcome (PO) and the corresponding Program Educational Objective (PEO). The justification demonstrates how the intended graduate attributes support the broader educational objectives of the program.

- **PO1 to PEO1:** The knowledge of mathematics, natural sciences, engineering fundamentals, and specialization equips students with the foundational skills necessary to analyze, design, and develop engineering solutions, aligning with PEO1's focus on applying computing and analytical approaches.
- **PO2 to PEO1:** Identifying and analyzing complex engineering problems using the first principles of mathematics, natural sciences, and engineering sciences ensures that students can tackle real-world problems effectively, supporting PEO1.
- **PO3 to PEO1:** Designing solutions for complex engineering problems with consideration for public health, safety, and environmental concerns ensures that graduates can create practical and responsible solutions, which is a core aspect of PEO1.
- **PO4 to PEO1:** Conducting investigations using research-based knowledge and methods aligns to engage in cutting-edge research and development, as outlined in PEO1.

- **PO5 to PEO1:** Using modern tools and techniques is essential for developing effective engineering solutions, directly supporting the analytical and solution-development aspects of PEO1.
- **PO6 to PEO2:** Applying contextual knowledge to assess societal, health, safety, legal, and cultural issues ensures that graduates understand their professional responsibilities and the impact of their work on society, which is central to PEO2.
- **PO7 to PEO2:** Understanding sustainability and the environmental impact of engineering work aligns with the ethical considerations and responsibilities emphasized in PEO2.
- **PO8 to PEO2:** Commitment to professional ethics and responsibilities is fundamental to demonstrating professionalism and ethical values, as outlined in PEO2.
- **PO9 to PEO4:** Functioning effectively as an individual and in diverse teams supports the development of teamwork and communication skills, which are crucial for interacting with people from various educational and cultural backgrounds, as per PEO4.
- **PO10 to PEO4:** Effective communication skills are necessary for successful interaction with diverse audiences and for teamwork, aligning with PEO4.
- **PO11 to PEO4:** Knowledge of project management and finance, and the ability to apply these principles, supports working effectively in multidisciplinary teams and managing projects, as required by PEO4.
- **PO12 to PEO3:** Recognizing the need for and engaging in lifelong learning ensures that graduates remain motivated for continuous professional development, aligning with PEO3.

8.2 Course Outcomes (COs)

The department maintains a structured process for the design, review, and continuous improvement of course outcomes and course delivery. All course instructors prepare draft course outlines, course contents, assessment plans, and CO–PO mappings in accordance with the Academic Council (AC)-approved OBE curriculum and departmental syllabus. To ensure consistency across multiple sections of the same course, designated course coordinators oversee the harmonization of course contents, assessments, and outcome mappings.

At the beginning of each semester, the proposed course outlines, including Course Outcomes (COs), assessment schemes, and CO–PO mappings, are reviewed within the department. The Departmental Committee (DC), comprising all faculty members, evaluates the proposed course materials, attainment data, and recommendations from previous semesters. The committee also reviews Continuous Quality Improvement (CQI) findings, instructor feedback, student feedback, alumni feedback, employer feedback, and course-level attainment results to identify areas requiring improvement.

Recommendations involving course content revisions, syllabus updates, course repositioning, credit-hour modifications, or curriculum enhancements are subsequently reviewed by the Departmental Syllabus Committee to ensure alignment with the program mission, Program Outcomes (POs), and Program Educational Objectives (PEOs). Significant curricular changes are forwarded through the institutional approval process and are ultimately ratified by the Academic Council before implementation.

Based on the review outcomes and CQI findings from previous cycles, necessary improvements are incorporated into the course outlines, assessment strategies, and CO definitions. The COs are finalized after departmental approval, ensuring that each course remains aligned with the OBE framework and supports the continuous improvement objectives of the program.

8.2.1 Statements of COs

Course Outcomes (COs) are specific, measurable statements that describe the knowledge, skills, and competencies students are expected to acquire upon successful completion of a course. For each core course offered by the program, the CO statements are developed in alignment with the approved OBE curriculum and are mapped to the relevant Program Outcomes (POs) and learning domains. These mappings provide the basis for course-level assessment, attainment measurement, and subsequent evaluation of PO attainment. The delivery methods, learning activities, and

assessment tools are also identified to ensure that each CO can be effectively taught, assessed, and continuously improved. The CO statements, corresponding PO mappings, learning domains, delivery methods, and assessment tools for the core courses are presented in Table 8.3.

Table 8.3: Statement of COs

HUM 4441: Engineering Ethics					
COs	**CO Description**	**PO**	**Learning Domains**	**Delivery Methods and Activities**	**Assessment Tools**
CO1	To develop an understanding of the moral and ethical principles that govern the practice and decision making of engineering.	PO6	C2	Lecture, Group Discussion, Problem Solving	Mid, Final
CO2	To examine the social and cultural implications along with professional responsibility and accountability.	PO8	C4	Lecture, Group Discussion, Problem Solving	Mid, Final
CO3	To provide a framework for ethical analysis and decision-making in engineering contexts.	PO8	C3	Lecture, Group Discussion, Problem Solving	Mid, Final
CO4	To analyze and apply the ethical implications of data-driven technologies and issues of data privacy, security, and ownership.	PO6	C4	Lecture, Group Discussion, Problem Solving	Mid, Final
CO5	To understand the role of data-driven technologies in creating and perpetuating social biases and discrimination.	PO6	C2	Lecture, Group Discussion, Problem Solving	Mid, Final
CSE 4513: Software Engineering and OOD					
COs	**CO Description**	**PO**	**Learning Domains**	**Delivery Methods and Activities**	**Assessment Tools**
CO1	**Explain** the fundamentals of the software development process and requirement engineering.	PO1, PO2	C2	Lecture, Problem Solving	Mid, Final
CO2	**Apply** appropriate design knowledge to model large SW systems.	PO2	C3	Lecture, Problem Solving	Mid, Final
CO3	**Demonstrate** an ability to use the techniques and tools to release and maintain quality software.	PO1, PO2	C3	Lecture, Problem Solving	Mid, Final
CSE 4711: Artificial Intelligence					
COs	**CO Description**	**PO**	**Learning Domains**	**Delivery Methods and Activities**	**Assessment Tools**
CO1	**Apply** general artificial intelligence techniques for common problem types	PO1	C3	Lecture, Demonstration	Mid, Final
CO2	**Formulate** various problem scenarios as environments and agents with proper justification	PO2	C5	Lecture, Demonstration	Mid, Final
CO3	**Recommend** solutions to real-life artificial intelligence problems by using the existing techniques	PO3	C6	Lecture, Demonstration	Mid, Final
CSE 4712: Artificial Intelligence Lab					

COs	CO Description	PO	Learning Domains	Delivery Methods and Activities	Assessment Tools
CO1	**Implement** and **analyze** a variety of algorithms to solve complex navigation and strategy problems.	PO5	C3, C4	Group Discussion, Demonstration, Problem Solving	Lab Viva, Lab Task, Lab Report
CO2	**Formulate** complex, real-world scenarios as specific AI paradigms.	PO2	C5	Group Discussion, Demonstration, Problem Solving	Lab Viva, Lab Task, Lab Report
CO3	**Design, test,** and **evaluate** intelligent agents that make rational decisions under uncertainty using modern programming tools and computational frameworks.	PO4	C6, P5	Group Discussion, Demonstration, Problem Solving	Lab Viva, Lab Task, Lab Report
CO4	**Communicate** the analysis, **design** choices, and performance insights of implemented AI solutions through structured technical reports.	PO10	P7, C5	Group Discussion, Demonstration, Problem Solving	Lab Viva, Lab Task, Lab Report
CSE 4610: Design Project					
COs	**CO Description**	**PO**	**Learning Domains**	**Delivery Methods and Activities**	**Assessment Tools**
CO1	**Identify** research problems and implement innovative solutions by exploring, analyzing, and applying state-of-the-art technologies, frameworks, tools, and algorithms to address complex real-world engineering challenges.	PO1, PO2, PO3, PO4, PO5	C4	Group Discussion, Demonstration, Project	Presentation, Project
CO2	Illustrate effective teamwork, professional ethics, and responsibility by applying project management principles to plan, execute, and deliver a complex software system within defined constraints.	PO8, PO9, PO11	C3, A4	Group Discussion, Demonstration, Project	Presentation, Project
CO3	Evaluate and **design** computing solutions considering ethical, societal, and environmental implications, ensuring responsible and sustainable engineering practices.	PO6, PO7, PO8	C5, A3	Group Discussion, Demonstration, Project	Presentation, Project
CO4	**Communicate** project outcomes through a well-structured research report and formal presentation.	PO10	C3, A3	Group Discussion, Demonstration, Project	Presentation, Project
CO5	**Develop** a strong foundation for future research in the same or related domains.	PO12	C2, A2	Group Discussion, Demonstration, Project	Presentation, Project
CSE 4700: Project/Thesis					
COs	**CO Description**	**PO**	**Learning Domains**	**Delivery Methods and Activities**	**Assessment Tools**
CO1	**Identify** and **formulate** a significant engineering research problem by reviewing relevant literature and analyzing existing solutions.	PO1, PO2, PO4	C4, A4	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report

CO2	**Design** or **propose** appropriate methodologies, models, or systems to address the identified problem.	PO3	C6, A5, P5	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO3	**Conduct** experiments, simulations, or data-driven investigations using modern tools and software to evaluate proposed solutions and interpret results effectively.	PO4, PO5	C4, A4, P4	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO5	**Prepare** a comprehensive technical report and **deliver** an effective oral presentation communicating research findings suitable for both technical and non-technical audiences.	PO10	C5, P5, A7	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO6	**Work** independently and collaboratively while applying project management principles and **demonstrating** a commitment to continuous learning and improvement.	PO9, PO11, PO12	C6, P5, A7	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CSE 4800: Project/Thesis					
COs	**CO Description**	**PO**	**Learning Domains**	**Delivery Methods and Activities**	**Assessment Tools**
CO2	**Design** or **propose** appropriate methodologies, models, or systems to address the identified problem.	PO3	C6, A5, P5	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO3	**Conduct** experiments, simulations, or data-driven investigations using modern tools and software to evaluate proposed solutions and interpret results effectively.	PO4, PO5	C4, A4, P4	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO4	**Demonstrate** ethical responsibility, professionalism, and awareness of standard ethical guidelines and responsibilities in the research process, considering social and environmental context.	PO6, PO7, PO8	C6, P6	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO5	**Prepare** a comprehensive technical report and **deliver** an effective oral presentation communicating research findings suitable for both technical and non-technical audiences.	PO10	C5, P5, A7	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report
CO6	**Work** independently and collaboratively while applying project management principles and **demonstrating** a commitment to continuous learning and improvement.	PO9, PO11, PO12	C6, P5, A7	Group Discussion, Demonstration, Problem Solving, Report Writing	Presentation, Report

8.2.2 Relationship between COs and POs

The relationship between Course Outcomes (COs) and Program Outcomes (POs) is established to ensure that the learning outcomes of individual courses collectively contribute to the attainment of the graduate attributes expected from the program. Each CO is mapped to one or more relevant POs based on the knowledge, skills, and professional competencies addressed through the course contents, learning activities, and assessment tools. This mapping provides a clear link between course-level assessment and program-level outcome attainment.

As shown in Table 8.4, the core courses contribute to different POs according to their nature and learning objectives. HUM 4441 emphasizes professional responsibility, social context, and ethics by addressing PO6 and PO8. CSE 4513 and CSE 4711 primarily support technical outcomes such as engineering knowledge, problem analysis, and design/development of solutions through PO1, PO2, and PO3. CSE 4712 extends these outcomes through laboratory-based activities involving problem analysis, investigation, design, and communication, thereby contributing to PO2, PO3, PO4, and PO10. The project and thesis-based courses, including CSE 4610, CSE 4700, and CSE 4800, provide broader coverage of the POs by integrating design, investigation, modern tool usage, societal and ethical considerations, teamwork, project management, and life-long learning. This distribution ensures that both technical and professional outcomes are addressed through appropriate courses and assessment mechanisms. Table 8.4 presents the CO–PO mapping for the core courses considered for outcome assessment.

Table 8.4: Relationship between COs and POs

Course code (Title)	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
HUM 4441 (Engineering Ethics)	CO1						✓						
	CO2								✓				
	CO3								✓				
	CO4						✓						
	CO5						✓						
CSE 4513 (Software Engineering)	CO1	✓	✓										
	CO2		✓										
	CO3	✓	✓										
CSE 4711 (Artificial Intelligence)	CO1	✓											
	CO2		✓										
	CO3			✓									
CSE 4712 (Artificial Intelligence Lab)	CO1				✓								
	CO2		✓										
	CO3			✓									
	CO4										✓		
CSE 4610 (Design Project)	CO1	✓	✓	✓	✓	✓							
	CO2								✓	✓		✓	
	CO3						✓	✓	✓				
	CO4										✓		
	CO5												✓
CSE 4700 (Project/Thesis)	CO1	✓	✓		✓								
	CO2			✓									
	CO3				✓	✓							
	CO5						✓						
	CO6									✓		✓	✓
CSE 4800 (Project/Thesis)	CO2			✓									
	CO3				✓	✓							
	CO4						✓	✓	✓				
	CO5												
	CO6									✓		✓	✓

8.3 Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities

The B.Sc. in CSE curriculum is designed to ensure that students acquire the required Knowledge Profile (K), experience the characteristics of Complex Engineering Problems (P), and engage in Complex Engineering Activities (A) throughout the program. The Knowledge Profile defines the depth and breadth of knowledge expected from engineering graduates, including natural sciences, mathematics, engineering fundamentals, specialist computing knowledge, engineering design, engineering practice, societal and ethical understanding, and engagement with research literature. Complex Engineering Problems refer to problems that require in-depth engineering knowledge, analysis of conflicting requirements, abstract and original thinking, consideration of unfamiliar issues, awareness of codes and standards, stakeholder involvement, and the handling of interdependent sub-problems. Complex Engineering Activities refer to professional activities that involve diverse resources, significant interaction among technical and non-technical issues, innovation, societal and environmental consequences, and application beyond previous experience.

The curriculum addresses these attributes through theory, laboratory, design, and project/thesis-based courses. As summarized in Table 8.9, foundation and theory-oriented courses such as HUM 4441, CSE 4513, and CSE 4711 emphasize ethical, societal, mathematical, engineering fundamental, and specialist knowledge components. Laboratory and project-oriented courses such as CSE 4712, CSE 4610, CSE 4700, and CSE 4800 provide broader coverage of complex problem and activity attributes by requiring students to implement solutions, conduct investigations, use modern tools, evaluate alternatives, communicate results, work in teams, manage project resources, and consider societal, environmental, ethical, and life-long learning aspects. This structured mapping demonstrates that the program progressively exposes students to the expected K, P, and A attributes and supports the attainment of relevant Program Outcomes (POs).

Table 8.5: Knowledge Profile Attributes

Attribute	Characteristics of Knowledge Profile
WK1: Theory-based natural sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline.
WK2: Conceptually-based mathematics, numerical analysis, statistics, and formal aspects of computer and information science	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline.
WK3: Theory-based engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4: Forefront engineering specialist knowledge for practice	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of discipline.
WK5: Engineering design	Knowledge that supports engineering design in a practice area.
WK6: Engineering practice (technology)	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7: Comprehension of engineering in society	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
WK8: Research literature	Engagement with selected knowledge in the research literature of the discipline

Table 8.8 summarizes the coverage of Knowledge Profile (WK1–WK8), Complex Engineering Problems (CEP1–CEP7), and Complex Engineering Activities (CEA1–CEA5) across the core courses considered in this section. A ✓ indicates that the corresponding attribute is addressed by the course.

Table 8.6: Complex Engineering Problem Attributes

Attribute	Characteristics of Complex Engineering Problem
CEP1: Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6, or WK8 which allows a fundamental-based, first principles analytical approach.
CEP2: Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues
CEP3: Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
CEP4: Familiarity of issues	Involve infrequently encountered issues
CEP5: Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering
CEP6: Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs
CEP7: Interdependence	Are high level problems including many component parts or sub-problems

Table 8.7: Complex Engineering Activities Attributes

Attribute	Characteristics of Complex Engineering Activities
CEA1: Range of resources	Involve the use of diverse resources (for this purpose, resources include people, money, equipment, materials, information and technologies)
CEA2: Level of interaction	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues
CEA3: Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways
CEA4: Consequences for society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
CEA5: Familiarity	Can extend beyond previous experiences by applying principle-based approaches

Table 8.8: Attainment summary of WK, CEP, and CEA attributes across core courses

Course	WK								CEP							CEA				
	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	1	2	3	4	5
HUM 4441				✓			✓													
CSE 4513	✓	✓	✓	✓	✓															
CSE 4711	✓	✓	✓	✓	✓				✓		✓	✓			✓					
CSE 4712		✓	✓	✓	✓				✓		✓	✓			✓		✓	✓		✓
CSE 4610	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓			
CSE 4700	✓	✓	✓	✓			✓	✓	✓				✓			✓	✓	✓	✓	
CSE 4800	✓	✓	✓	✓			✓	✓	✓				✓				✓	✓	✓	
Attainment of KPA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 8.9 presents the detailed course-wise mapping and justification for how the K, P, and A attributes are covered across each course listed in Section 1.2.1.

Table 8.9: Mapping of Core Courses to PO, WK, CEP, and CEA

Courses	COs	POs	WK	CEP	CEA	Justification
HUM 4441 (Engineering Ethics)	CO1	PO6	7	-	-	Focuses on moral and ethical principles governing engineering practice and decision-making, which are central elements of professional responsibility and the societal role of engineers.
	CO2	PO8	7	-	-	Examines social, cultural, and professional implications of engineering activities, directly addressing the societal and professional dimensions of engineering practice.
	CO3	PO8	7	-	-	Develops ethical analysis and decision-making frameworks that support responsible engineering practice and adherence to professional ethical standards.
	CO4	PO6	7, 4	-	-	Addresses ethical issues related to data privacy, security, and ownership, requiring both an understanding of societal and ethical responsibilities (WK7) and knowledge of data-driven technologies (WK4)
	CO5	PO6	7, 4	-	-	Explores technological causes of bias and discrimination in data-driven systems, requiring ethical awareness of societal impacts (WK7) and specialist knowledge of modern computing technologies (WK4)
CSE 4513 (Software Engineering and OOD)	CO1	PO1, PO2	1, 2	-	-	WK1 because software engineering principles are built upon scientific and systematic approaches to problem solving. WK2 because requirement engineering involves logical reasoning, abstraction, modeling, and formal representation of system requirements.
	CO2	PO2	3, 4	-	-	WK3 because object-oriented design relies on engineering fundamentals such as abstraction, modularity, encapsulation, and decomposition. WK4 because software design methodologies and modeling techniques constitute specialized software engineering knowledge.
	CO3	PO1, PO2	5	-	-	WK5 directly supports engineering design and implementation practices, including software quality assurance, maintenance, and deployment concepts.
CSE 4711 (Artificial Intelligence)	CO1	PO1	1, 2, 3	1	-	WK1 Artificial intelligence is grounded in scientific principles related to reasoning, search, learning, and intelligent behavior. WK2 AI techniques require logical reasoning, mathematical representations, probability, and formal computational models for problem solving. WK3 Students apply engineering fundamentals such as state-space representation, search algorithms, knowledge representation, and problem-solving methodologies. CEP1 Applying AI techniques requires in-depth engineering knowledge of algorithms, search methods, and intelligent systems, enabling a fundamentals-based analytical approach to solving computational problems.

Courses	COs	POs	WK	CEP	CEA	Justification
	CO2	PO2	3, 4	1, 3, 4	-	WK3 Problem formulation relies on engineering fundamentals such as state-space representation, search algorithms, and problem-solving methodologies. WK4 Agent-based problem solving and environment modeling constitute specialist knowledge within artificial intelligence. CEP1 Agent formulation requires specialized AI knowledge and analytical reasoning. CEP3 Formulating environments and agents often has no single correct representation and requires abstract thinking and model construction. CEP4 Many AI applications involve unfamiliar or diverse real-world scenarios that must be translated into formal problem representations.
	CO3	PO3	4, 5	1, 5, 7	-	WK4 Selecting appropriate AI techniques requires specialist knowledge of artificial intelligence methods and their capabilities. WK5 Recommending solutions involve design-oriented thinking where suitable AI approaches are chosen to satisfy application requirements. CEP1 Solution recommendation requires substantial AI knowledge and analytical reasoning. CEP5 Real-world AI problems frequently permit multiple solution approaches, requiring evaluation and justification of alternatives. CEP7 Real-life AI applications often consist of multiple interconnected components such as data representation, reasoning, learning, and decision-making, requiring consideration of several sub-problems within a larger system.
CSE 4712 (Artificial Intelligence Lab)	CO1	PO5	2, 3, 4	1, 3, 5	3, 5, 6	WK2 Application of mathematical and computational concepts used in search, optimization, heuristics, and state-space exploration. WK3 Application of engineering fundamentals related to algorithm design and analysis. WK4 Use of specialized artificial intelligence techniques such as search algorithms, game-playing strategies, and intelligent problem-solving methods. CEP1 Implementation and analysis require in-depth AI and computing knowledge. CEP3 Different algorithms may produce different outcomes. CEA3 Students implement AI algorithms using programming tools and apply AI principles in practical situations. CEA5 Lab tasks require applying learned concepts to problems beyond worked classroom examples. CEA6 Different approaches require critical evaluation and selection.

Courses	COs	POs	WK	CEP	CEA	Justification
	CO2	PO2	3, 4	1, 3, 4	2, 5	WK3 Uses engineering fundamentals of abstraction, decomposition, and modeling. WK4 Requires specialist AI knowledge to represent problems as search spaces, reasoning, or learning tasks. CEP1 Problem formulation requires substantial AI knowledge. CEP3 There is often no single correct representation of a problem. CEP4 Many real-world scenarios encountered in the lab are unfamiliar and require new problem representations. CEA2 Students resolve conflicts between real-world complexity and computational models. CEA5 Students extend theoretical concepts to previously unseen practical scenarios.
	CO3	PO4	4, 5	1, 3, 7	2, 5, 6	WK4 Requires specialist knowledge of intelligent agents, reasoning, and decision-making. WK5 Directly involves engineering design through agent architecture, behavior, testing, and evaluation. CEP1 Agent development requires advanced AI knowledge. CEP3 Multiple design alternatives exist and require evaluation. CEP7 Intelligent agents contain several interacting subsystems such as perception, reasoning, decision-making, and evaluation. CEA2 Students manage interactions among algorithms, data, frameworks, and performance requirements. CEA5 Agent design involves creative application of AI principles and computational tools. CEA6 Students design and evaluate solutions that extend beyond standard examples covered in lectures.
	CO4	PO10	4, 5	1, 2, 7	2, 5, 6	WK4 Communication involves specialist AI concepts and terminology. WK5 Students justify engineering design choices and evaluation outcomes. CEP7 Reporting requires integrating results from multiple components of an AI solution and presenting them coherently. CEP2 Students explain trade-offs, design decisions, and interactions among system components. CEP5 Technical reporting involves applying engineering communication practices to document and evaluate novel implementations. CEA6 Students develop and present original AI solutions that extend beyond example applications.

Courses	COs	POs	WK	CEP	CEA	Justification
CSE 4610 (Design Project)	CO1	PO1, PO2, PO3, PO4, PO5	1, 2, 3, 4, 5, 8	1, 2, 3, 4, 7	1, 2	WK1 Application of scientific principles and systematic knowledge to investigate and understand real-world engineering problems. WK2 Use of mathematical, computational, and analytical techniques to model, analyze, and evaluate engineering solutions. WK3 Application of engineering fundamentals in problem identification, analysis, and solution development. WK4 Utilization of specialist computing knowledge, modern technologies, frameworks, algorithms, and tools relevant to the project domain. WK5 Application of engineering design principles to develop innovative and effective solutions for complex engineering challenges. WK8 Engagement with current research literature and state-of-the-art developments to identify research problems and justify solution approaches. CEP1 Project execution requires in-depth engineering knowledge spanning multiple knowledge domains. CEP2 Real-world projects involve conflicting technical, performance, resource, and stakeholder requirements. CEP3 Research-oriented design problems require abstraction, innovation, and evaluation of alternative solution strategies. CEP4 Students frequently encounter unfamiliar technologies, problem domains, and implementation challenges. CEP7 Complex projects consist of multiple interrelated components requiring integration and system-level thinking. CEA1 Project development involves the coordinated use of diverse resources including software tools, computing platforms, information sources, and team members. CEA2 Students resolve challenges arising from interactions among technical, engineering, and practical project constraints.
	CO2	PO8, PO9, PO11	6, 7	1, 2, 6	1, 2	WK6 Application of engineering practices, development methodologies, project management techniques, and professional workflows. WK7 Understanding professional ethics, responsibility, teamwork, leadership, and the role of engineers in society. CEP2 Team projects require balancing technical objectives, project constraints, schedules, and stakeholder expectations. CEP6 Successful project execution involves collaboration among diverse stakeholders with varying responsibilities and requirements. CEA1 Project management requires allocation and coordination of people, resources, schedules, and technologies. CEA2 Team members must resolve conflicts and interactions among technical, managerial, and organizational issues throughout the project lifecycle.

Courses	COs	POs	WK	CEP	CEA	Justification
	CO3	PO6, PO7, PO8	5, 7	1, 2, 6	2	WK5 Application of design knowledge to evaluate alternative solutions and incorporate societal and environmental considerations. WK7 Understanding ethical responsibilities and the societal, environmental, and sustainability impacts of engineering solutions. CEP2 Evaluation of computing solutions requires balancing technical effectiveness with ethical, social, and environmental concerns. CEP6 Engineering decisions affect multiple stakeholder groups with diverse and sometimes conflicting needs. CEA2 Students analyze and resolve interactions among technical objectives, ethical considerations, sustainability concerns, and societal impacts.
	CO4	PO10	4, 8	1, 7	2	WK4 Communication requires specialist knowledge of the project's technical concepts, methodologies, implementation details, and outcomes. WK8 Preparation of reports and presentations requires synthesis of project findings and supporting research literature. CEP7 Effective communication requires integration and presentation of multiple project components, results, analyses, and conclusions. CEA2 Students communicate and justify technical decisions, trade-offs, methodologies, and project outcomes arising from complex engineering activities.
	CO5	PO12	4, 8	1, 4	2	WK4 Continued learning requires understanding emerging technologies, tools, frameworks, and specialist knowledge within the project domain. WK8 Students engage with research literature and advanced knowledge sources to identify future research directions. CEP4 Research-oriented exploration exposes students to unfamiliar problems, technologies, and evolving areas of knowledge. CEA2 Students critically evaluate existing solutions and identify opportunities for future investigation and improvement through principled engineering approaches.
CSE 4700 (Project Thesis)	CO1	PO1, PO2, PO4	2, 3, 4, 8	1, 2, 4	1, 4	WK2 (Mathematics) – required for analytical reasoning in problem formulation. WK3 (Engineering fundamentals) – needed to understand existing systems. WK4 (Specialist knowledge) – necessary for identifying domain-specific gaps. WK8 (Research literature) – directly required for literature review. CEP1 (Depth of knowledge) – research problems require deep domain understanding. CEP2 (Conflicting requirements) – identifying gaps involves trade-offs. CEP4 (Familiarity) – problems are generally non-routine. CEA1 (Resources) – use of literature, datasets, and tools. CEA4 (Societal/environmental consequences) – focus on significant, impactful problems.

Courses	COs	POs	WK	CEP	CEA	Justification
	CO2	PO3	3, 4	1, 3, 5	3	WK3 (Engineering fundamentals) – foundation for solution development. WK4 (Specialist knowledge) – advanced knowledge for system design. CEP3 (Depth of analysis) – design problems require original modeling and have no direct solutions. CEP3 (Innovation) – design involves creative application of knowledge.
	CO3	PO4, PO5	3, 4, 8	1, 2, 3, 4	2, 3	WK3 (Engineering fundamentals) – for designing experiments. WK4 (Specialist knowledge) – for using domain-specific tools and methods. WK8 (Research literature) – for interpreting results in context. CEP2 (Conflicting requirements) – experiments involve multiple interacting variables. CEP3 (Depth of analysis) – requires analytical reasoning. CEP4 (Familiarity) – often involves unfamiliar or exploratory scenarios. CEA2 (Interaction) – managing interactions between variables and tools. CEA3 (Innovation) – experiment design is non-routine and creative.
	CO5	PO10	1, 2, 8	1, 7	2, 4	WK1 (Natural science) – understanding the technical context. WK2 (Mathematics/statistics) – presenting data accurately. WK8 (Research literature) – structuring reports based on prior work. CEP7 (Interdependence) – communication involves explaining interconnected components. CEA2 (Interaction) – communication with different stakeholders. CEA4 (Societal impact) – addressing both technical and non-technical audiences.
	CO6	PO9, PO11, PO12	2, 7	1, 6, 7	1, 2	WK2 (Mathematics/analytical thinking) – planning and structured problem-solving. WK7 (Professional knowledge) – teamwork and responsibility. CEP6 (Stakeholders) – teamwork involves multiple individuals. CEP7 (Interdependence) – tasks within a project are interconnected. CEA1 (Resources) – managing time, tools, and personnel. CEA2 (Interaction) – coordination and collaboration.
CSE 4800 (Project Thesis)	CO2	PO3	3, 4	1, 3, 5	3	WK3 (Engineering fundamentals) – foundation for solution development. WK4 (Specialist knowledge) – advanced knowledge for system design. CEP3 (Depth of analysis) – design problems require original modeling and have no direct solutions. CEP3 (Innovation) – design involves creative application of knowledge.

Courses	COs	POs	WK	CEP	CEA	Justification
	CO3	PO4, PO5	3, 4, 8	1, 2, 3, 4	2, 3	WK3 (Engineering fundamentals) – for designing experiments. WK4 (Specialist knowledge) – for using domain-specific tools and methods. WK8 (Research literature) – for interpreting results in context. CEP2 (Conflicting requirements) – experiments involve multiple interacting variables. CEP3 (Depth of analysis) – requires analytical reasoning. CEP4 (Familiarity) – often involves unfamiliar or exploratory scenarios. CEA2 (Interaction) – managing interactions between variables and tools. CEA3 (Innovation) – experiment design is non-routine and creative.
	CO4	PO8, PO6, PO7	8, 7	1, 2, 6	3, 4	WK7 (Ethics, society, sustainability) – addresses ethical responsibility and societal/environmental considerations in research practice. WK8 (Research literature) – supports application of ethical standards through engagement with research processes and guidelines. CEP6 (Stakeholders) – involves consideration of diverse stakeholders in ethical and societal contexts. CEP2 (Conflicting requirements) – reflects balancing ethical responsibilities with research objectives and societal impact. CEA4 (Societal consequences) – focuses on evaluating the impact of engineering and research outcomes on society and environment. CEA3 (Innovation/engineering judgment) – involves applying ethical principles and professional judgment in research analysis and reporting.
	CO5	PO10	1, 2, 8	1, 7	2, 4	WK1 (Natural science) – understanding the technical context. WK2 (Mathematics/statistics) – presenting data accurately. WK8 (Research literature) – structuring reports based on prior work. CEP7 (Interdependence) – communication involves explaining interconnected components. CEA2 (Interaction) – communication with different stakeholders. CEA4 (Societal impact) – addressing both technical and non-technical audiences.
	CO6	PO9, PO11, PO12	2, 7	1, 6, 7	1, 2	WK2 (Mathematics/analytical thinking) – planning and structured problem-solving. WK7 (Professional knowledge) – teamwork and responsibility. CEP6 (Stakeholders) – teamwork involves multiple individuals. CEP7 (Interdependence) – tasks within a project are interconnected. CEA1 (Resources) – managing time, tools, and personnel. CEA2 (Interaction) – coordination and collaboration.

8.4 Assessment of COs

The course outcomes must be measurable in an outcome-based education system. The CSE program at Islamic University of Technology uses direct methods for the assessment of COs through direct examinations and other measurable performance parameters, such as course projects, laboratory tasks and examinations.

Assessment Tools and Rubrics

The COs are assessed course-wise. Table 8.10 shows the tools used to measure a student's performance or skills in a course using direct methods. Appropriate assessment tools are chosen based on the contents, course outcomes, and CO-to-PO mapping of individual courses. Our courses are categorized into theory courses of 2, 3, and 4 credits, and laboratory courses of 0.75, 1.0, and 1.5 credits. For theory courses, we can evaluate COs using quizzes, assignments, mid-term, and semester-final examinations. For laboratory courses, COs are assessed through lab tasks, lab viva and exams. Course projects are used in CO assessment where project components such as report writing and project presentations are included. CO mapping for quizzes and assignments is optional and may be recorded under "Others" based on the course assessment design. We also use class participation for grading students, but it is not included in CO assessment calculations.

Table 8.10: Assessment Tools for direct assessment methods

Assessment Tools	Explanation
Quizzes and assignments	The students are evaluated using quizzes and assignments designed to assess students' cognitive abilities and problem-solving skills and to meet the learning objectives of an individual course.
Mid-term exam	The students are evaluated through the mid-term examination with CO-mapped questions to assess attainment at individual and cohort levels.
Semester-final exam	The students are evaluated through the semester-final examination with CO-mapped questions to assess attainment at individual and cohort levels.
Lab tasks, Viva exams	The students are assigned several tasks in the laboratory based on theoretical concepts to improve psychomotor skills. They are assessed based on task completion, laboratory performance, Question Answering Session during Viva, reports, and Lab examinations.
Course project	The students are assigned course projects to enhance leadership and communication skills. They are assessed based on theoretical and practical design, report writing, presentation, and demonstration.

Student performance is evaluated using both quantitative marks and qualitative observations based on predefined assessment criteria and rubrics. These rubrics establish clear performance benchmarks ranging from Unsatisfactory to Excellent and are used to ensure consistency and objectivity in the evaluation process. The complete set of rubrics used across all courses is provided in Criterion 6. As an illustrative example, Table 8.11 presents the rubric used in CSE 4700 (Project/Thesis). In this course, students are evaluated by both their supervisor and the pre-defense/defense examination committee. The assessment criteria include regular meetings, teamwork, report quality, presentation skills, question-answering ability, and depth of understanding, with marks allocated according to the level of performance demonstrated. These rubric-based evaluations form the basis for determining student attainment and contribute to the overall assessment of the corresponding course outcomes and program outcomes.

Table 8.11: Rubrics for CSE 4700 (Project/Thesis) evaluation

Evaluator	Category	Unsat.	Insuf.	Adeq.	Sat.	Exc.
Supervisor	Regular Meeting	2	4	6	8	10
	Team Work	2	4	6	8	10
	Report	8	16	24	32	40
Pre-defense /Defense	Presentation (10)	2	4	6	8	10
	Q&A (10)	2	4	6	8	10
	Depth of Understanding (20)	4	8	12	16	20

CO Assessment Process

The Department of Computer Science and Engineering (CSE) at the Islamic University of Technology follows a systematic Outcome-Based Education (OBE) assessment framework to evaluate the attainment of Course Outcomes (COs) and Program Outcomes (POs). The framework ensures that student performance is assessed consistently using predefined assessment tools, CO-PO mappings, attainment thresholds, and cohort-based evaluation criteria. The resulting attainment values are used to measure the extent to which students achieve the intended learning outcomes of individual courses and, ultimately, the overall program outcomes.

The Department of Computer Science and Engineering (CSE) at the Islamic University of Technology has established the Course Outcome (CO) attainment threshold at 50%. Upon completion of grading for a course, the cohort size is determined by excluding students who receive an 'F' (Fail) grade from the total number of students enrolled in that course during the semester. A student belonging to the cohort is considered to have attained a particular CO if he or she achieves at least 50% of the marks allocated to that CO. Furthermore, the cohort-level attainment threshold is also set at 50%. Therefore, a CO is considered attained if at least 50% of the students in the cohort achieve the minimum required score of 50% for that CO. The assessment process is explained in detail through the following subsections.

Methodology for CO Assessment

This subsection describes the calculation procedure used to determine CO attainment at both individual-student and cohort levels. The methodology combines mark-based individual assessment with cohort-based attainment decisions so that course outcomes can be evaluated consistently across courses.

CO Attainment by an Individual Student: At the individual student level, the obtained marks are considered. If the ratio of the obtained mark in a particular CO of an individual student in the cohort set is at least 50% of the total allocated mark of that CO, then that CO is said to be attained by that individual student.

$$X = \frac{\sum \text{obtained marks in } CO_i}{\sum \text{total marks in } CO_i} \tag{8.1}$$

In Equation 8.1, X is the percentage of marks an individual student obtained for a particular CO. Now, if $X \geq 50\%$, then CO_i is considered to be attained by an individual student.

CO Attainment by a Cohort: The term "cohort" is interchangeably used with the term "class/batch" usually under one course instructor. If at least 50% of the students in the cohort of a class achieve a particular CO, then that CO is considered to be attained at the class level.

$$Y = \frac{\text{Number of students who achieved } CO_i \text{ in the class}}{\text{Cohort size of the class}} \tag{8.2}$$

Here, in Equation 8.2, Y denotes the percentage of students in the cohort who achieved at least 50% of the marks allocated to a particular CO. In other words, Y represents the percentage of students who attained that Course Outcome. Now, if $Y \geq 50\%$, then CO_i is attained for the overall class or course.

Explanation of CO Assessment Process with Example

This subsection illustrates the CO assessment methodology through a representative course example. The example shows how CO-mapped marks are allocated, how individual student attainment is calculated, and how cohort-level attainment decisions are derived.

An example of the CO assessment and attainment process, both for an individual student and at course level, of the core course CSE 4711 (Artificial Intelligence) is provided below.

Table 8.12 represents the overall mark distribution or assessment scheme for CSE 4711. The course has three COs, and CO-mapped marks are assigned for mid-term and semester-final examinations, while non-CO components are recorded separately under "Others". The weightage of COs is assigned in the "Total CO" row of the table.

At the end of the semester, the marks obtained by each student for the individual Course Outcomes (COs) are compiled, as illustrated in Table 8.13. These marks are then converted into percentages using the CO assessment formula

Table 8.12: Marks (%) distribution for CSE 4711 assessment tools

Assessment Items	CO1	CO2	CO3	Others	Total
Class Participation	-	-	-	10	10
Quizzes and Assignments	-	-	-	15	15
Mid-term	9.00	7.67	8.33	-	25
Semester-final	19.67	11.33	19.00	-	50
Total CO	28.67	19.00	27.33	25	100

Table 8.13: Obtained marks of selected students in each CO for CSE 4711

SL	Student ID	Name	CO1	CO2	CO3	CO Total	CO %
1	200041101	Abu Hena Shadid	48.5	40.0	11.0	99.5	44.22
2	200041105	Tawsif Tashwar Dipto	69.5	34.0	31.5	135.0	60.00
3	200041120	Khawaja Ishmam Arian	80.0	36.5	39.5	156.0	69.33

presented earlier in this section. An example of the resulting CO assessment for student ID 200041120 is shown in Table 8.14. Based on the established attainment criterion, a student is considered to have attained a CO if the corresponding assessment percentage is at least 50%. The attainment status of each CO is subsequently determined using this threshold.

Table 8.14: Sample of CO assessment for an individual student in CSE 4711

COs	Total	Obtained	CO achievement (Threshold value is 50%)	Attainment Status
CO1	118	80.0	CO1 = (80.0 / 118) × 100% = 67.80%	Attained
CO2	51	36.5	CO2 = (36.5 / 51) × 100% = 71.57%	Attained
CO3	56	39.5	CO3 = (39.5 / 56) × 100% = 70.54%	Attained

Finally, cohort-level CO attainment is determined by examining the percentage of students who attained each CO. According to the department's attainment policy, a CO is considered attained if at least 50% of the students in the cohort achieve a minimum of 50% of the marks allocated to that CO. The resulting cohort-level attainment for CSE 4711 is presented in Table 8.15.

As shown in the Figure 8.1, CO1 was attained by 40 out of 68 students, corresponding to an attainment percentage of 58.8%, which exceeds the departmental threshold and is therefore considered attained. Similarly, CO2 was attained by 50 out of 68 students (73.5%), indicating a strong level of achievement across the cohort. In contrast, CO3 was attained by only 27 out of 68 students, resulting in an attainment percentage of 39.7%, which falls below the required threshold of 50%. Consequently, CO3 is classified as not attained. Based on these results, CO1 and CO2 are considered achieved at the cohort level, whereas CO3 requires further attention and possible corrective actions CQI process to improve student performance in future offerings of the course.

Table 8.16 presents a summary of the cohort-level CO attainment results for all assessed courses. The results indicate that the majority of the Course Outcomes (COs) across the curriculum have been successfully attained, exceeding the departmental attainment threshold of 50%.

Among the assessed courses, HUM 4441 demonstrated strong attainment across all mapped COs, with attainment percentages ranging from 77.94% to 95.59%. Similarly, CSE 4610, CSE 4700, and CSE 4800 achieved consistently high attainment levels, with all reported COs exceeding 82%. In particular, CSE 4700 exhibited the highest overall attainment, with CO attainment percentages ranging from 91.18% to 94.12%.

CSE 4513 and CSE 4712 also showed satisfactory performance, as all assessed COs surpassed the 50% attainment threshold. In CSE 4513, attainment percentages ranged from 64.71% to 86.76%, while in CSE 4712 they ranged from 60.29% to 89.71%.

However, the attainment results for CSE 4711 indicate that not all Course Outcomes were achieved. While CO2 at-

Table 8.15: CO attainment by a Cohort for CSE 4711

COs	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CO1	40/68	58.8%	Attained
CO2	50/68	73.5%	Attained
CO3	27/68	39.7%	Not Attained

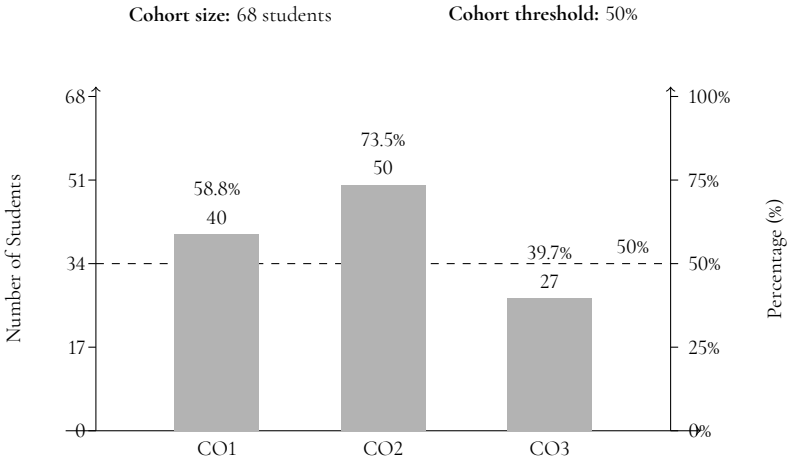


Figure 8.1: Bar chart of Cohort attainment summary of CSE 4711

Table 8.16: Summary of CO attainment for Core courses

Course	CO1	CO2	CO3	CO4	CO5	CO6
HUM 4441	91.18%	95.59%	77.94%	89.71%	85.29%	–
CSE 4513	86.76%	64.71%	66.18%	–	–	–
CSE 4711	36.76%	61.76%	20.59%	–	–	–
CSE 4712	82.35%	89.71%	60.29%	85.29%	–	–
CSE 4610	88.24%	88.24%	88.24%	82.35%	88.24%	–
CSE 4700	92.65%	92.65%	91.18%	–	94.12%	94.12%
CSE 4800	–	89.71%	91.18%	89.71%	91.18%	89.71%

tained a satisfactory level of 61.76%, CO1 and CO3 recorded attainment percentages of 36.76% and 20.59%, respectively, which are below the departmental threshold. Consequently, CO1 and CO3 of CSE 4711 are considered not attained and may require further investigation and corrective actions in future course offerings.

Overall, the attainment results demonstrate that the majority of Course Outcomes across the assessed courses have been achieved by the respective cohorts, providing evidence that students are successfully meeting the intended learning outcomes of the program.

8.5 Attainment of POs required by the BAETE

The Department of Computer Science & Engineering (CSE) at IUT defined COs along with the statements for core courses. Accordingly, the COs of the assessed courses are mapped with the 12 POs. From the contributions of PO attainment, a mathematically sound methodology was agreed upon for Course Level PO and Graduating Level PO measurements. Therefore, two types of PO measurements are defined as follows:

- **Course Level PO Measurement:** Individual PO Attainment and Cohort PO Attainment at the course level.
- **Graduating Level PO Measurement:** Individual PO Attainment ustilizing Course Level PO Measurement of the graduating cohort.

Methodology for Course Level PO Measurement

This subsection presents the procedure for measuring PO attainment within an individual course. It explains how CO-mapped assessment marks are accumulated under the relevant POs and how individual and cohort attainment decisions are made at the course level. Similar to the previous section, CSE 4711 is used as an illustrative example to demonstrate the Program Outcome (PO) attainment calculation process. The following subsections present the required data and calculations in a step-by-step manner to clearly show how the attainment values are determined.

The term "Course Level" refers to the class/course-section level in the context of PO. For every class, the obtained marks of a particular PO are accumulated from the marks of the COs, as per a fixed CO-to-PO mapping. Since binary mapping is used, the weights are either 0 or 1 at the course level.

$$X = \frac{\sum \text{obtained marks mapped to } PO_i}{\sum \text{allocated marks mapped to } PO_i} \tag{8.3}$$

In Equation 8.3, X denotes percentage of marks a student attained for a particular PO. Now, if $X \geq 40\%$, then PO_i is considered to be attained by an individual student.

Table 8.17: CO–PO mapping scheme for CSE 4711

Course	CO	Mapped POs
CSE 4711	CO1	PO1
CSE 4711	CO2	PO2
CSE 4711	CO3	PO3

For example, CSE 4711 (Artificial Intelligence) has the CO-to-PO mapping shown in Table 8.17. The obtained marks of student ID 200041120 are shown in Table 8.18, and the corresponding PO attainment calculation is shown in Table 8.19.

Table 8.18: CO marks used for CSE 4711 PO calculation

CO	Allocated Mark	Obtained Mark	Percentage
CO1	118	80.0	67.80%
CO2	51	36.5	71.57%
CO3	56	39.5	70.54%

Table 8.19: Individual PO attainment calculation for CSE 4711

PO	Allocated Mark	Obtained Mark	Percentage	Attainement Status
PO1	118	80.0	67.80%	Attained
PO2	51	36.5	71.57%	Attained
PO3	56	39.5	70.54%	Attained

For the cohort-set assessment of that class, if at least 50% of the students from the cohort-set achieve a particular PO, then that PO is considered to be achieved by that class.

$$Y = \frac{\text{Number of students who achieved } PO_i \text{ in the class}}{\text{Cohort size of the class}} \tag{8.4}$$

Now, if $Y \geq 50\%$, then PO_i is considered to be attained by that class.

Table 8.20: Cohort PO attainment for CSE 4711

PO	Number of Students Attained	Percentage of Students Attainment	Attainement Status
PO1	50/68	73.53%	Attained
PO2	54/68	79.41%	Attained
PO3	33/68	48.53%	Not Attained

For the CSE 4711 course-level example, the PO contribution records and the corresponding course-level PO attainment summary are shown in Figure 8.2 and Figure 8.3.

Course Code: CSE 4711 **Section:** 01
Course Title: Artificial Intelligence | PO Contribution at course level
Credit: 3.0

SL	Student ID	Name	PO1 (118)	PO2 (51)	PO3 (56)	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1	200041101	Abu Hena Shadid	48.5	40.0	11.0	–	–	–	–	–	–	–	–	–
2	200041105	Tawsif Tashwar Dipto	69.5	34.0	31.5	–	–	–	–	–	–	–	–	–
3	200041106	Asif Abrar	87.0	39.0	32.5	–	–	–	–	–	–	–	–	–
4	200041108	Jarin Tasnim Hridy	53.5	36.0	20.5	–	–	–	–	–	–	–	–	–
5	200041109	Fahib Abrar	38.5	22.0	11.0	–	–	–	–	–	–	–	–	–

Figure 8.2: PO contribution at course level of CSE 4711

Methodology for Graduating Level PO Measurement

This subsection explains the graduating-level PO measurement process, which evaluates the cumulative attainment of Program Outcomes (POs) for the graduating cohort. The method utilizes culminating courses and their contributions to determine whether graduates have achieved the expected program outcomes.

Similar to the course-level assessment, the graduating cohort is considered the cohort-set for graduating-level PO measurement. Each student has already been individually assessed for the attainment of PO_i through the course-level PO measurement process described in the previous subsection. Therefore, at the graduating level, the objective is to determine whether PO_i has been attained through any of the courses mapped to it. If at least one mapped course achieves PO_i, then PO_i is considered attained for that student. The distribution of Program Outcomes (POs) across the culminating courses, as well as the corresponding PO mappings for each culminating course, are presented in Table 8.21 and Table 8.22, respectively.

The graduating-level attainment of PO_i is depicted in Equation 8.5. In Equation 8.5, PO_i^j denote the attainment status of Program Outcome PO_i through course j. C_i is the set of courses mapped to PO_i, and $\bigvee$ represents the logical OR operation. Thus, PO_i is attained if any of the mapped courses successfully attains PO_i.

$$PO_i = \bigvee_{j \in C_i} PO_i^j \tag{8.5}$$

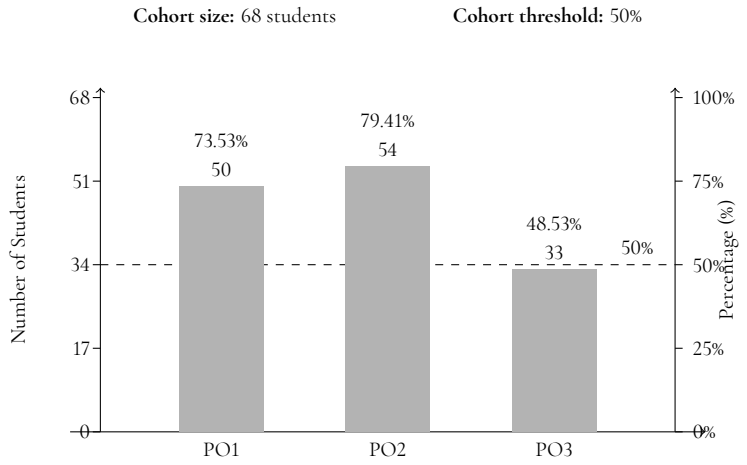


Figure 8.3: Bar chart of course-level PO attainment summary of CSE 4711

Table 8.21: Distribution of culmination courses among different POs

PO	Courses
PO1	CSE 4513, CSE 4711, CSE 4610, CSE 4700
PO2	CSE 4513, CSE 4711, CSE 4712, CSE 4610, CSE 4700
PO3	CSE 4711, CSE 4610, CSE 4700, CSE 4800
PO4	CSE 4712, CSE 4610, CSE 4700, CSE 4800
PO5	CSE 4712, CSE 4610, CSE 4700, CSE 4800
PO6	HUM 4441, CSE 4610, CSE 4800
PO7	CSE 4610, CSE 4800
PO8	HUM 4441, CSE 4610, CSE 4800
PO9	CSE 4610, CSE 4700, CSE 4800
PO10	CSE 4712, CSE 4610, CSE 4700, CSE 4800
PO11	CSE 4700, CSE 4800
PO12	CSE 4610, CSE 4700, CSE 4800

Table 8.22: List of culminating courses for Graduating PO assessment

Culminating Course	PO Mapping
HUM 4441 (Engineering Ethics)	PO6, PO8
CSE 4513 (Software Engineering and OOD)	PO1, PO2
CSE 4711 (Artificial Intelligence)	PO1, PO2, PO3
CSE 4712 (Artificial Intelligence Lab)	PO2, PO4, PO5, PO10
CSE 4610 (Design Project)	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO12
CSE 4700 (Project/Thesis)	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PO12
CSE 4800 (Project/Thesis)	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12

8.5.1 Attainment of PO1: Engineering Knowledge

The assessment and measurement of PO1 (Engineering Knowledge) are conducted through direct methods in CSE 4513, CSE 4711, CSE 4610, and CSE 4700. These courses require students to apply knowledge of mathematics, natural sciences, engineering fundamentals, and engineering specialization.

As shown in Table 8.23, all courses mapped to PO1 have successfully attained the outcome, as each course achieved an attainment percentage of at least 50%. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO1 has been achieved at the graduating level. Furthermore, Figure 8.4 provides a graphical representation of the attainment results for better visualization and interpretation.

Table 8.23: PO1 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CSE 4513	64/68	94.12%	Attained
CSE 4711	50/68	73.53%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4700	66/68	97.06%	Attained

In CSE 4513, students demonstrate their ability to apply software engineering principles and object-oriented design concepts, while in CSE 4711, they apply fundamental artificial intelligence techniques to solve computational problems. For both courses, the Mid-Term and Semester Final examinations were used as assessment tools. Course instructors designed assessment questions specifically intended to evaluate PO1 attainment. These questions underwent moderation to ensure their alignment with PO1 before being administered to students. Based on the marks obtained in the relevant questions, the course-level attainment of PO1 was calculated, and the results are presented in Table 8.23.

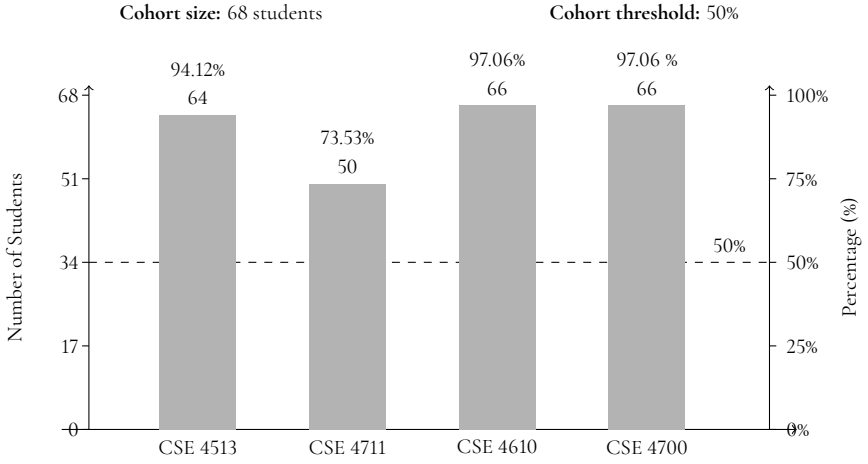


Figure 8.4: PO1 attainment by mapped courses

In CSE 4610 and CSE 4700, students demonstrate the application of engineering knowledge through design projects and project/thesis activities. Reports were used as the assessment tool in both courses to evaluate PO1 attainment. In each course, PO1 is mapped to CO1, which requires students to apply engineering knowledge to identify, formulate, and analyze research problems. The respective project or thesis supervisors evaluated the extent to which students successfully applied engineering principles in defining and formulating a valid research problem and assigned marks accordingly. These assessment results were subsequently used to calculate the course-level attainment of PO1, as shown in Table 8.23.

8.5.2 Attainment of PO2: Problem Analysis

The attainment of PO2 (Problem Analysis) is assessed and measured through direct methods in CSE 4513, CSE 4711, CSE 4712, CSE 4610, and CSE 4700. These courses require students to identify, formulate, research literature, and analyze complex computer science and engineering problems using first principles of mathematics, natural sciences, and engineering sciences.

As shown in Table 8.24, all five courses mapped to PO2 have successfully attained the outcome, with each course exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO2 has been achieved at the graduating level. Furthermore, Figure 8.5 provides a graphical representation of the attainment results for better visualization and interpretation.

Table 8.24: PO2 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainment Status
CSE 4513	61/68	89.71%	Attained
CSE 4711	54/68	79.41%	Attained
CSE 4712	61/68	89.71%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4700	66/68	97.06%	Attained

In CSE 4513, PO2 is mapped to CO1, CO2, and CO3. CO1 requires students to explain the fundamentals of the software development process and requirement engineering, CO2 requires students to apply appropriate design knowledge to model large software systems, and CO3 requires students to demonstrate the use of techniques and tools to release and maintain quality software. In CSE 4711, PO2 is mapped to CO2, which requires students to formulate complex, real-world scenarios as specific AI paradigms with proper justification. For both courses, Mid-Term and Semester Final examinations were used as assessment tools. Course instructors designed assessment questions specifically intended to evaluate PO2 attainment. For the assessment criteria and rubric details, refer to the respective course outlines. Based on the marks obtained in the relevant questions, the course-level attainment of PO2 was calculated, and the results are presented in Table 8.24.

In CSE 4712, PO2 is mapped to CO2, which requires students to formulate complex, real-world scenarios as specific AI paradigms through hands-on lab tasks. Lab tasks were used as the assessment tool. For the assessment criteria and rubric details, refer to the CSE 4712 course outline. Based on the marks obtained in the relevant lab tasks, the course-level attainment of PO2 was calculated, and the results are presented in Table 8.24.

In CSE 4610 and CSE 4700, PO2 is mapped to CO1 in both courses. In CSE 4610, CO1 requires students to identify research problems and implement innovative solutions by exploring, analyzing, and applying state-of-the-art technologies, frameworks, and algorithms. In CSE 4700, CO1 requires students to identify and formulate a significant engineering research problem by reviewing relevant literature and analyzing existing solutions. Reports and presentations were used as assessment tools in both courses. The respective supervisors evaluated the extent to which students successfully analyzed the problem and formulated a well-grounded research direction and assigned marks accordingly. These assessment results were subsequently used to calculate the course-level attainment of PO2, as shown in Table 8.24.

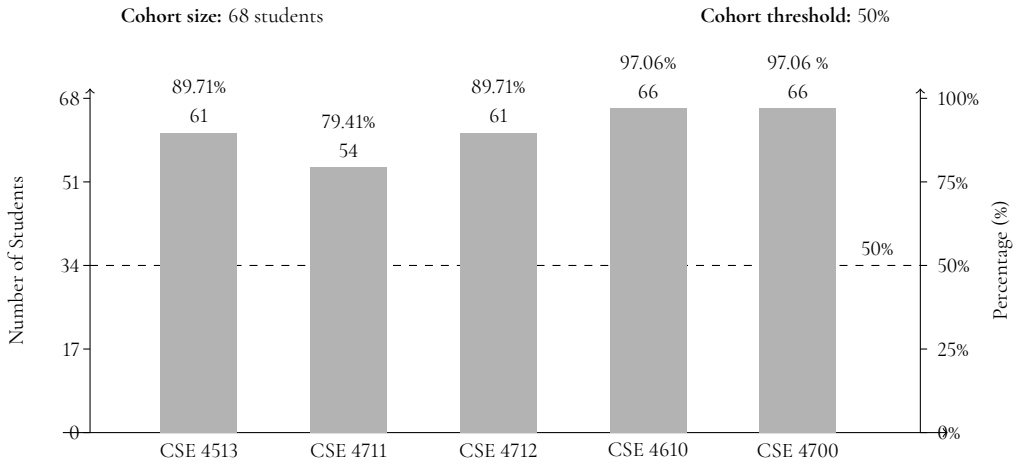


Figure 8.5: PO2 attainment by mapped courses

8.5.3 Attainment of PO3: Design/Development of Solutions

The attainment of PO3 (Design/Development of Solutions) is assessed and measured through direct methods in CSE 4711, CSE 4610, CSE 4700, and CSE 4800. These courses require students to design solutions for complex engineering problems and develop systems, components, or processes that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental factors.

As shown in Table 8.25, three of the four courses mapped to PO3 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. CSE 4711 recorded a cohort attainment of 48.53%, which falls marginally below the threshold and is therefore classified as not attained. However, since the graduating-level PO attainment condition defined in Equation 8.5 requires at least one mapped course to attain the PO, and three courses have done so, PO3 is considered attained at the graduating level. The below-threshold result for CSE 4711 has been noted and will be addressed through the Continuous Quality Improvement (CQI) process in future course offerings. Figure 8.6 provides a graphical representation of the attainment results.

Table 8.25: PO3 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainment Status
CSE 4711	33/68	48.53%	Not Attained
CSE 4610	66/68	97.06%	Attained
CSE 4700	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

In CSE 4711, PO3 is mapped to CO3, which requires students to recommend solutions to real-life artificial intelligence problems by selecting and applying existing AI techniques. Mid-Term and Semester Final examinations were used as assessment tools. For the assessment criteria and rubric details, refer to the CSE 4711 course outline. Based on the marks obtained in the relevant questions, the course-level attainment of PO3 was calculated, and the results are presented in Table 8.25.

In CSE 4610, PO3 is mapped to CO1, which requires students to identify research problems and implement innovative solutions by exploring, analyzing, and applying state-of-the-art technologies, frameworks, tools, and algorithms to address complex real-world engineering challenges. Reports and presentations were used as assessment tools. The respective supervisors evaluated the design quality and assigned marks accordingly.

In CSE 4700 and CSE 4800, PO3 is mapped to CO2 in both courses, which requires students to design or propose appropriate methodologies, models, or systems to address the identified research problem. Reports were used as the assessment tool in both courses. The Proposed Methodology chapter of the thesis report is marked out of 10 by the supervisor, using a five-level rubric: Unsatisfactory (2 marks) for a methodology disconnected from the problem; Insufficient (4) for a poorly reasoned methodology; Adequate (6) for a logically sound methodology with basic justification; Satisfactory (8) for a well-structured methodology with clear justification and appropriate diagrams; and Excellent (10) for an original, comprehensive, and thoroughly justified methodology grounded in relevant prior work. The respective supervisors evaluated the methodology and assigned marks accordingly. These assessment results were subsequently used to calculate the course-level attainment of PO3, as shown in Table 8.25.

8.5.4 Attainment of PO4: Investigation

The attainment of PO4 (Investigation) is assessed and measured through direct methods in CSE 4712, CSE 4610, CSE 4700, and CSE 4800. These courses require students to conduct investigations of complex engineering problems using research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

As shown in Table 8.26, all four courses mapped to PO4 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO4 has been achieved at the graduating level. Furthermore, Figure 8.7 provides a graphical representation of the attainment results for better visualization and interpretation.

In CSE 4712, PO4 is mapped to CO3, which requires students to design, test, and evaluate intelligent agents that make rational decisions under uncertainty using modern programming tools and computational frameworks. Lab tasks were

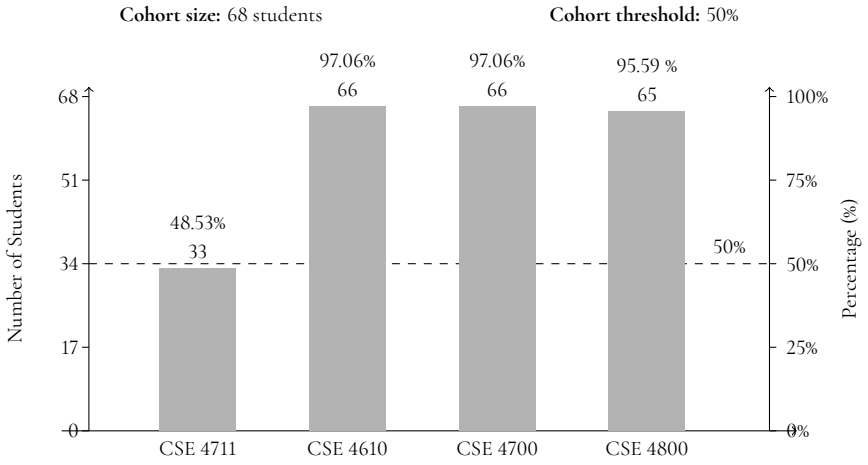


Figure 8.6: PO3 attainment by mapped courses

Table 8.26: PO4 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainment Status
CSE 4712	49/68	72.06%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4700	65/68	95.59%	Attained
CSE 4800	65/68	95.59%	Attained

used as the assessment tool. For the assessment criteria and rubric details, refer to the CSE 4712 course outline. Based on the marks obtained in the relevant lab tasks, the course-level attainment of PO4 was calculated, and the results are presented in Table 8.26.

In CSE 4610, PO4 is mapped to CO1, which requires students to identify research problems and implement innovative solutions, including conducting experiments or evaluations on the implemented system and interpreting results in the context of the research objectives. Reports and presentations were used as assessment tools. The respective supervisors evaluated the quality of investigation and assigned marks accordingly.

In CSE 4700 and CSE 4800, PO4 is mapped to CO3 in both courses, which requires students to conduct experiments, simulations, or data-driven investigations using modern tools and software to evaluate proposed solutions and interpret results effectively. Reports were used as the assessment tool in both courses. The Results and Discussion chapter of the thesis report is marked out of 10 by the supervisor, using a five-level rubric: Unsatisfactory (2 marks) for missing or incoherent evaluation; Insufficient (4) for results presented without proper discussion; Adequate (6) for a structured evaluation with basic interpretation; Satisfactory (8) for meaningful experiments with clear discussion and comparison against baselines; and Excellent (10) for a thorough, well-designed investigation with insightful analysis and a repository link with implementation details. The respective supervisors evaluated the quality of the investigation and assigned marks accordingly. These assessment results were subsequently used to calculate the course-level attainment of PO4, as shown in Table 8.26.

8.5.5 Attainment of PO5: Modern Tool Usage

The attainment of PO5 (Modern Tool Usage) is assessed and measured through direct methods in CSE 4712, CSE 4610, CSE 4700, and CSE 4800. These courses require students to create, select, and apply appropriate techniques, resources, and modern engineering and IT tools—including prediction and modelling—to complex engineering problems, with an understanding of their limitations.

As shown in Table 8.27, all four courses mapped to PO5 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO5 has been achieved at the graduating level. Furthermore, Figure 8.8 provides a graphical representation of the attainment results for better visualization and interpretation.

In CSE 4712, PO5 is mapped to CO1, which requires students to implement and analyze a variety of algorithms to

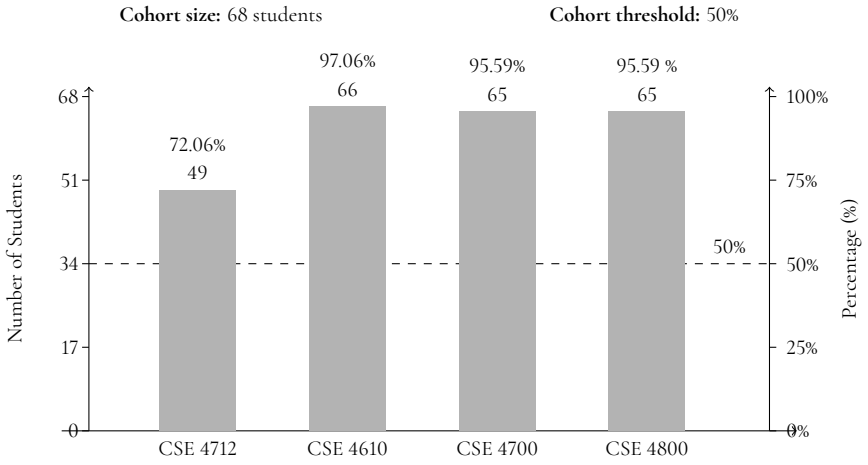


Figure 8.7: PO4 attainment by mapped courses

Table 8.27: PO5 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainment Status
CSE 4712	59/68	86.76%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4700	65/68	95.59%	Attained
CSE 4800	65/68	95.59%	Attained

solve complex navigation and strategy problems using modern programming tools and computational frameworks such as Python. Lab tasks were used as the assessment tool. For the assessment criteria and rubric details, refer to the CSE 4712 course outline. Based on the marks obtained in the relevant lab tasks, the course-level attainment of PO5 was calculated, and the results are presented in Table 8.27.

In CSE 4610, PO5 is mapped to CO1, which requires students to identify research problems and implement innovative solutions by exploring, analyzing, and applying state-of-the-art technologies, frameworks, tools, and algorithms to address complex real-world engineering challenges. Reports and presentations were used as assessment tools. The respective supervisors evaluated the appropriateness and effectiveness of tool usage and assigned marks accordingly.

In CSE 4700 and CSE 4800, PO5 is mapped to CO3 in both courses, which requires students to conduct experiments using modern tools and software and to reflect on their suitability and limitations in the context of the investigation. Reports were used as the assessment tool in both courses. The Results and Discussion chapter of the thesis report is marked out of 10 by the supervisor, using a five-level rubric: Unsatisfactory (2 marks) for missing or inappropriate tool usage; Insufficient (4) for tool usage that is present but poorly applied; Adequate (6) for correct use of appropriate tools with basic experimental evaluation; Satisfactory (8) for effective and well-justified tool selection with meaningful results; and Excellent (10) for sophisticated use of modern tools, well-designed experiments, and a repository link with full implementation details. The respective supervisors evaluated the appropriateness and effectiveness of tool usage and assigned marks accordingly. These assessment results were subsequently used to calculate the course-level attainment of PO5, as shown in Table 8.27.

8.5.6 Attainment of PO6: The Engineer and Society

The attainment of PO6 (The Engineer and Society) is assessed and measured through direct methods in HUM 4441, CSE 4610, and CSE 4800. These courses require students to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.

As shown in Table 8.28, all three courses mapped to PO6 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO6 has been achieved at the graduating level. Furthermore, Figure 8.9 provides a graphical representation of the attainment results for better visualization and interpretation.

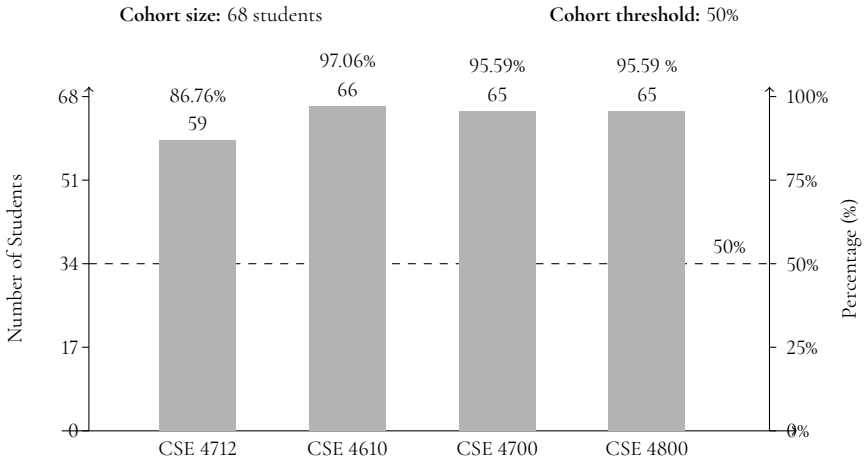


Figure 8.8: PO5 attainment by mapped courses

Table 8.28: PO6 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainment Status
HUM 4441	66/68	97.06%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

In HUM 4441, PO6 is mapped to CO1, CO4, and CO5. CO1 requires students to develop an understanding of the moral and ethical principles that govern engineering practice and decision-making. CO4 requires students to analyze and apply the ethical implications of data-driven technologies, including issues of data privacy, security, and ownership. CO5 requires students to understand the role of data-driven technologies in creating and perpetuating social biases and discrimination. Mid-Term and Semester Final examinations were used as assessment tools, with the Semester Final placing greater weight on CO4 and CO5. For the assessment criteria and rubric details, refer to the HUM 4441 course outline. Based on the marks obtained in the relevant questions, the course-level attainment of PO6 was calculated, and the results are presented in Table 8.28.

In CSE 4610 and CSE 4800, reports and presentations were used as assessment tools to evaluate PO6 attainment. In CSE 4610, PO6 is mapped to CO3, which requires students to evaluate and design computing solutions considering ethical, societal, and environmental implications, ensuring responsible and sustainable engineering practices. In CSE 4800, PO6 is mapped to CO4, which requires students to demonstrate ethical responsibility, professionalism, and awareness of standard ethical guidelines and responsibilities in the research process, considering social and environmental context. The respective supervisors evaluated the extent to which students successfully incorporated these considerations into their work and assigned marks accordingly. These assessment results were subsequently used to calculate the course-level attainment of PO6, as shown in Table 8.28.

8.5.7 Attainment of PO7: Environment and Sustainability

The attainment of PO7 (Environment and Sustainability) is assessed and measured through direct methods in CSE 4610 and CSE 4800. These courses require students to evaluate sustainability and the impact of professional engineering work in societal and environmental contexts.

As shown in Table 8.29, both courses mapped to PO7 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO7 has been achieved at the graduating level. Furthermore, Figure 8.10 provides a graphical representation of the attainment results for better visualization and interpretation.

For both CSE 4610 and CSE 4800, reports and regular presentations were used as assessment tools to evaluate PO7 attainment. In CSE 4610, PO7 is mapped to CO3, which requires students to evaluate and design computing solutions considering ethical, societal, and environmental implications, ensuring responsible and sustainable engineering practices. In CSE 4800, PO7 is mapped to CO3, which requires students to conduct experiments, simulations, or data-

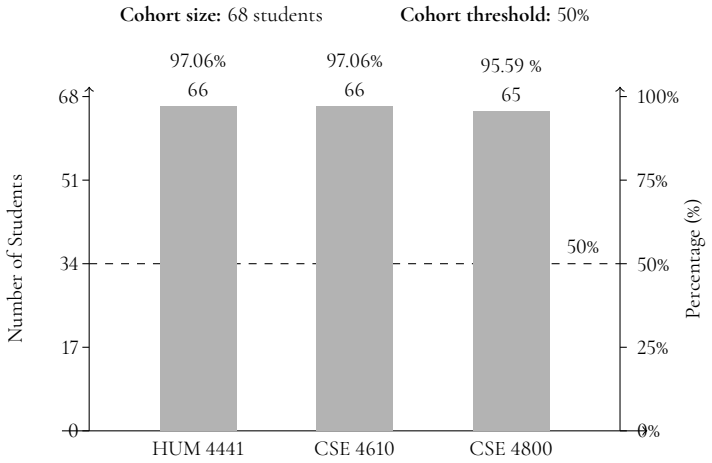


Figure 8.9: PO6 attainment by mapped courses

Table 8.29: PO7 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CSE 4610	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

driven investigations using modern tools and software to evaluate proposed solutions and interpret results effectively. The respective project or thesis supervisors evaluated the extent to which students successfully incorporated these considerations into their work and assigned marks accordingly. These assessment results were then used to calculate the course-level attainment of PO7, as shown in Table 8.29.

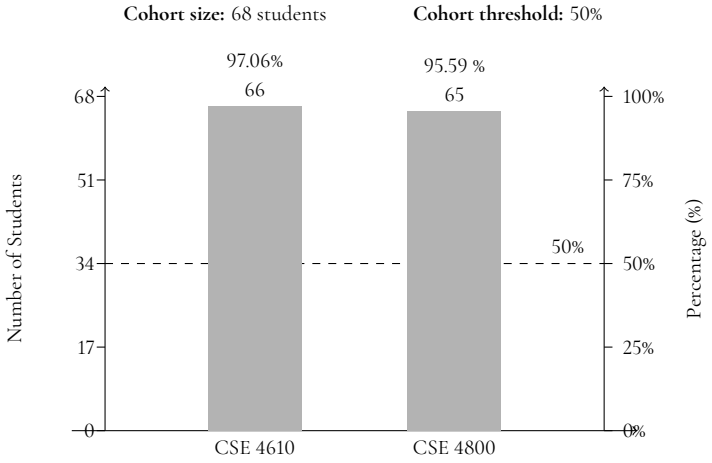


Figure 8.10: PO7 attainment by mapped courses

8.5.8 Attainment of PO8: Ethics

The attainment of PO8 (Ethics) is assessed and measured through direct methods in HUM 4441, CSE 4610, and CSE 4800. These courses require students to examine the social, cultural, and professional implications of engineering practice and to apply ethical principles in decision-making, project execution, and research conduct.

As shown in Table 8.30, all three courses mapped to PO8 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO8 has been achieved at the graduating level. Furthermore, Figure 8.11 provides a graphical representation of the attainment results for better visualization and interpretation.

HUM 4441 addresses professional responsibility and ethical decision-making directly. The Mid-Term and Semester Final examinations were used as assessment tools. In HUM 4441, PO8 is mapped to CO2 and CO3. CO2 examines

Table 8.30: PO8 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
HUM 4441	64/68	94.12%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

the social and cultural implications of engineering practice along with professional responsibility and accountability. CO3 provides a framework for ethical analysis and decision-making in engineering contexts. Based on the marks obtained in the relevant questions and activities, the course-level attainment of PO8 was calculated, and the results are presented in Table 8.30.

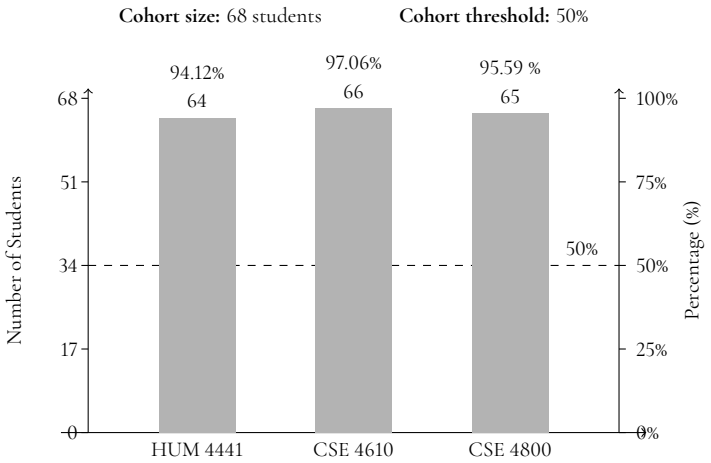


Figure 8.11: PO8 attainment by mapped courses

For both CSE 4610 and CSE 4800, reports and regular presentations were used as assessment tools to evaluate PO8 attainment. In CSE 4610, PO8 is mapped to CO2 and CO3. CO2 requires students to demonstrate effective teamwork, professional ethics, and responsibility by applying project management principles to plan, execute, and deliver a complex software system within defined constraints. CO3 requires students to evaluate and design computing solutions considering ethical, societal, and environmental implications, ensuring responsible and sustainable engineering practices. In CSE 4800, PO8 is mapped to CO4, which emphasizes ethical responsibility, professionalism, and awareness of standard ethical guidelines and responsibilities in the research process, considering social and environmental context. The respective project or thesis supervisors evaluated the extent to which students successfully incorporated these considerations into their work and assigned marks accordingly. These assessment results were then used to calculate the course-level attainment of PO8, as shown in Table 8.30.

8.5.9 Attainment of PO9: Individual Work and Teamwork

The attainment of PO9 (Individual Work and Teamwork) is assessed and measured through direct methods in CSE 4610, CSE 4700, and CSE 4800. These courses require students to function effectively as individuals and as members of teams in project and research settings, balancing independent contribution with collaborative delivery.

As shown in Table 8.31, all three courses mapped to PO9 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO9 has been achieved at the graduating level. Furthermore, Figure 8.12 provides a graphical representation of the attainment results for better visualization and interpretation.

In CSE 4610, PO9 is mapped to CO1 and CO2, where CO1 requires students to individually identify research problems and implement innovative solutions, while CO2 requires students to demonstrate effective teamwork and responsibility by applying project management principles to plan, execute, and deliver a complex software system. This individual-team balance was reflected in the assessment structure, where weekly meeting attendance and coding/artifact contribution captured individual input, while progress updates and the project report captured team-level collaboration. In CSE 4700 and CSE 4800, PO9 is mapped to CO6, which requires students to work independently and

Table 8.31: PO9 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CSE 4610	66/68	97.06%	Attained
CSE 4700	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

collaboratively while applying project management principles and demonstrating a commitment to continuous learning and improvement. In all three courses, reports and regular presentations were used as assessment tools, with the respective supervisors evaluating both individual contribution and team-based deliverables to determine the course-level attainment of PO9, as shown in Table 8.31.

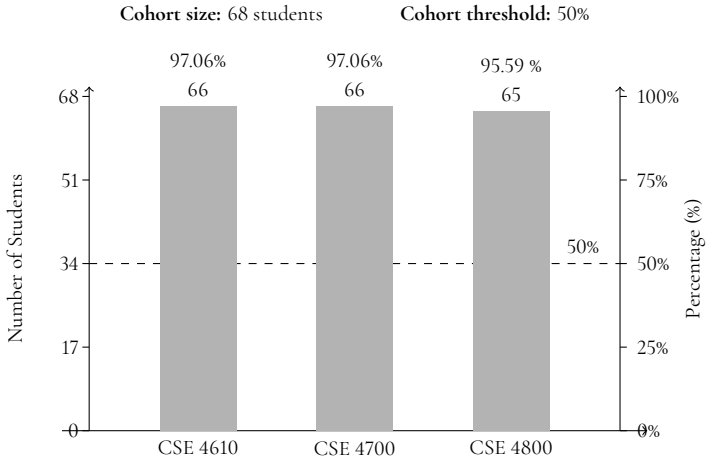


Figure 8.12: PO9 attainment by mapped courses

8.5.10 Attainment of PO10: Communication

The attainment of PO10 (Communication) is assessed and measured through direct methods in CSE 4712, CSE 4610, CSE 4700, and CSE 4800. These courses require students to communicate complex computer science and engineering activities through reports, presentations, demonstrations, and technical documentation.

As shown in Table 8.32, all four courses mapped to PO10 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO10 has been achieved at the graduating level. Furthermore, Figure 8.13 provides a graphical representation of the attainment results for better visualization and interpretation.

Table 8.32: PO10 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CSE 4712	65/68	95.59%	Attained
CSE 4610	66/68	97.06%	Attained
CSE 4700	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

CSE 4712 addresses communication through structured technical reporting. A final lab viva, along with a structural lab report for each lab, were used as assessment tools. In CSE 4712, PO10 is mapped to CO4, which requires students to communicate the analysis, design choices, and performance insights of implemented AI solutions through structured technical reports. Based on the marks obtained in the relevant questions and reports, the course-level attainment of PO10 was calculated, and the results are presented in Table 8.32.

In CSE 4610, CSE 4700, and CSE 4800, students communicate project/thesis objectives, methods, results, and conclusions to technical and non-technical audiences. Reports and presentations were used as the assessment tools in all three courses. In CSE 4610, PO10 is mapped to CO4, which requires students to communicate project outcomes

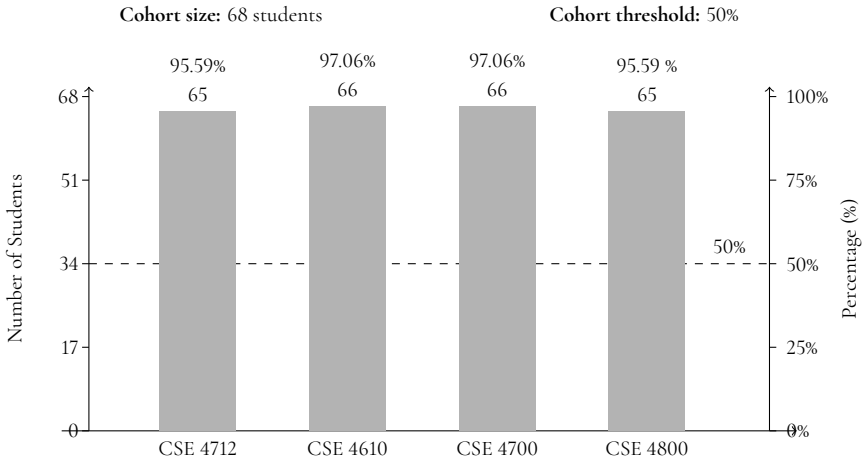


Figure 8.13: PO10 attainment by mapped courses

through a well-structured research report and formal presentation. In CSE 4700 and CSE 4800, PO10 is mapped to CO5, which requires students to prepare a comprehensive presentation and communicate research findings in a way that is suitable for both technical and non-technical audiences. The respective course instructors and supervisors evaluated the clarity, structure, and audience suitability of these communications and assigned marks accordingly. These assessment results were then used to calculate the course-level attainment of PO10, as shown in Table 8.32.

8.5.11 Attainment of PO11: Project Management and Finance

The attainment of PO11 (Project Management and Finance) is assessed and measured through direct methods in CSE 4700 and CSE 4800. These courses require students to demonstrate project management principles and apply them in project and thesis work. Students are assessed on planning, scheduling, resource management, milestone tracking, division of responsibilities, and execution of project/thesis activities within defined constraints.

As shown in Table 8.33, both courses mapped to PO11 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO11 has been achieved at the graduating level. Furthermore, Figure 8.14 provides a graphical representation of the attainment results for better visualization and interpretation.

Table 8.33: PO11 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CSE 4700	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

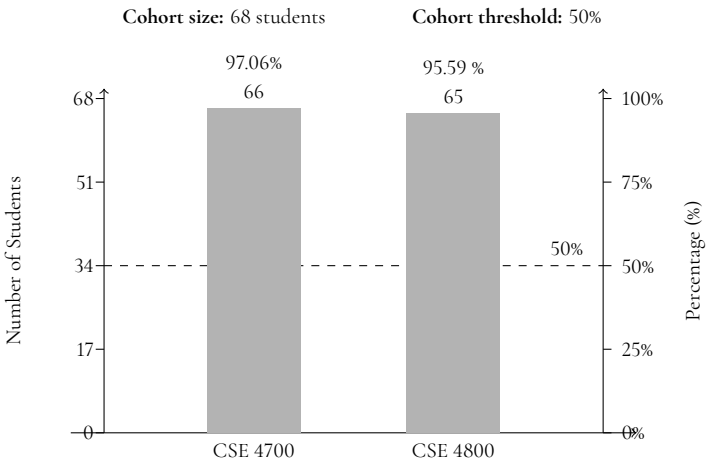


Figure 8.14: PO11 attainment by mapped courses

For both CSE 4700 and CSE 4800, reports and regular presentations were used as assessment tools to evaluate PO11 attainment. In both courses, PO11 is mapped to CO6, which requires students to work independently and collaboratively while applying project management principles and demonstrating a commitment to continuous learning and improvement. The respective project or thesis supervisors evaluated the extent to which students successfully applied these project management principles and assigned marks accordingly. These assessment results were then used to calculate the course-level attainment of PO11, as shown in Table 8.33.

8.5.12 Attainment of PO12: Life-Long Learning

The attainment of PO12 (Life-Long Learning) is assessed and measured through direct methods in CSE 4610, CSE 4700, and CSE 4800. These courses require students to demonstrate the preparation and ability to engage in independent and life-long learning in the context of technological change. Students demonstrate this outcome through independent study of new technologies, literature review, self-directed problem solving, adaptation to unfamiliar tools, and reflection on future improvement or research directions.

As shown in Table 8.34, all three courses mapped to PO12 have successfully attained the outcome, each exceeding the 50% cohort attainment threshold. Since this satisfies the condition defined in Equation 8.5, it can be concluded that PO12 has been achieved at the graduating level. Furthermore, Figure 8.15 provides a graphical representation of the attainment results for better visualization and interpretation.

Table 8.34: PO12 attainment by mapped courses

Courses	Number of Students Attained	Percentage of Students Attainment	Attainement Status
CSE 4610	66/68	97.06%	Attained
CSE 4700	66/68	97.06%	Attained
CSE 4800	65/68	95.59%	Attained

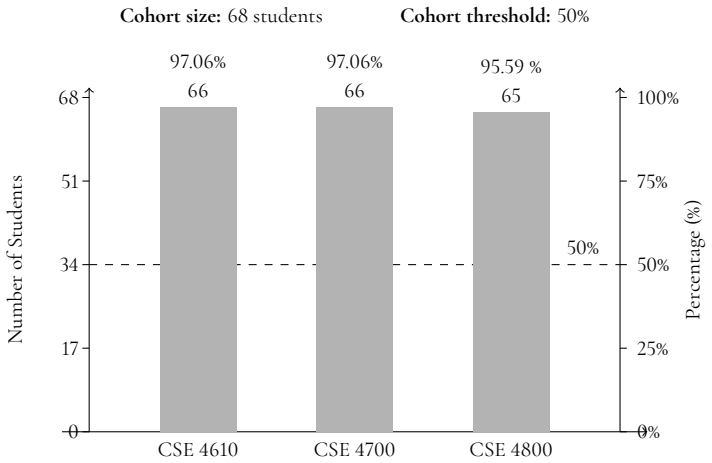


Figure 8.15: PO12 attainment by mapped courses

In CSE 4610, PO12 is mapped to CO5, which requires students to build a strong foundation for future research in the same or related domains. In CSE 4700 and CSE 4800, PO12 is mapped to CO6, which requires students to work independently and collaboratively while applying project management principles and demonstrating a commitment to continuous learning and improvement. Across all three courses, the assessment of life-long learning was embedded in the project/thesis process itself: students were required to explore unfamiliar tools, technologies, and literature beyond the taught curriculum, and to reflect on the broader applicability and future direction of their work in their reports and presentations. The respective supervisors evaluated this capacity for self-directed learning and assigned marks accordingly, and these results were used to calculate the course-level attainment of PO12, as shown in Table 8.34.

8.6 Achievement of additional POs

No additional Program Outcomes (POs) beyond the standard twelve POs have been defined by the program. Therefore, assessment and evaluation of additional POs are not applicable.